

```
import java.util.Scanner;

public class MatrixSubtraction {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        // Get the dimensions of both matrices
        System.out.println("Enter the number of rows in the matrices: ");
        int rows = scanner.nextInt();
        System.out.println("Enter the number of columns in the matrices: ");
        int columns = scanner.nextInt();

        // Initialize two matrices
        int[][] matrix1 = new int[rows][columns];
        int[][] matrix2 = new int[rows][columns];

        // Get user input for the first matrix
        System.out.println("Enter the elements of the first matrix (row-wise): ");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix1[i][j] = scanner.nextInt();
            }
        }

        // Get user input for the second matrix
        System.out.println("Enter the elements of the second matrix (row-wise): ");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix2[i][j] = scanner.nextInt();
            }
        }

        // Create a new matrix to store the difference
        int[][] differenceMatrix = new int[rows][columns];
```

```
// Subtract corresponding elements of each matrix to the difference matrix
```

```
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < columns; j++) {  
        differenceMatrix[i][j] = matrix1[i][j] - matrix2[i][j];  
    }  
}
```

```
// Print the original matrices and the difference matrix
```

```
System.out.println("Matrix 1:");  
printMatrix(matrix1);  
System.out.println("Matrix 2:");  
printMatrix(matrix2);  
System.out.println("Difference Matrix:");  
printMatrix(differenceMatrix);  
}
```

```
// Utility function to print a matrix
```

```
public static void printMatrix(int[][] matrix) {  
    for (int i = 0; i < matrix.length; i++) {  
        for (int j = 0; j < matrix[0].length; j++) {  
            System.out.print(matrix[i][j] + " ");  
        }  
        System.out.println();  
    }  
}
```